Infiltration

This section is to calculate the infiltration

```
#relevant libraries import
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os

#From https://www.abs.gov.au/census/find-census-data/quickstats/2021/SAL22315 the data of
#Meblurne subburbs population and area was optained

#creation of data frame based on population data
df = pd.read_csv('./melbourne_density.csv')


total_melbourne_population=df["Population"].mean()
total_melbourne_area=df["Area (km2)"].mean()

print("The total melbourne population is: " + str(total_melbourne_population))
print("The total melbourne area is: " + str(total_melbourne_area))

PE_domestic=PE_domestic #variable defined in previous python calculation notebook
print("The suburb under design population is: " + str(PE_domestic))
print(str(PE_domestic))
print(str(PE_commercial))

#estimation of under study suburb area based on area per population ratio

estimated_area=(total_melbourne_area/total_melbourne_population)*PE_domestic

print("The estimated area of the understudy suburb is: " + str(estimated_area) + 'km2')

#converting the estimated area into ha

estimated_area=estimated_area*100

print('The estimated area of the understudy suburb is: ' + str(estimated_area) + 'ha')

#from Metcalf and Eddy the infiltration for swewer can be estimated as 0.2 to 28 m3/ha/d
#lower value is taken as it is a new swewer

infiltration_ratio=0.2

infiltration=estimated_area*infiltration_ratio

print('The estimated infiltration is: ' + str(infiltration) +'m3/d')

#exporting data frame to latex
latex_table = df.to_latex(index=False,longtable=True, caption='Melbourne suburbs with area and population', label='tab:mel_subs')
with open("../document/melbourne_density.tex", "w") as f:
    f.write(latex_table)
```

```
The total melbourne population is: 11199.702380952382
The total melbourne area is: 18.649833333333337
The suburb under design population is: 340000
340000
28461
The estimated area of the understudy suburb is: 566.1706996890863km2
The estimated area of the understudy suburb is: 56617.06996890863ha
The estimated infiltration is: 11323.413993781727m3/d
```